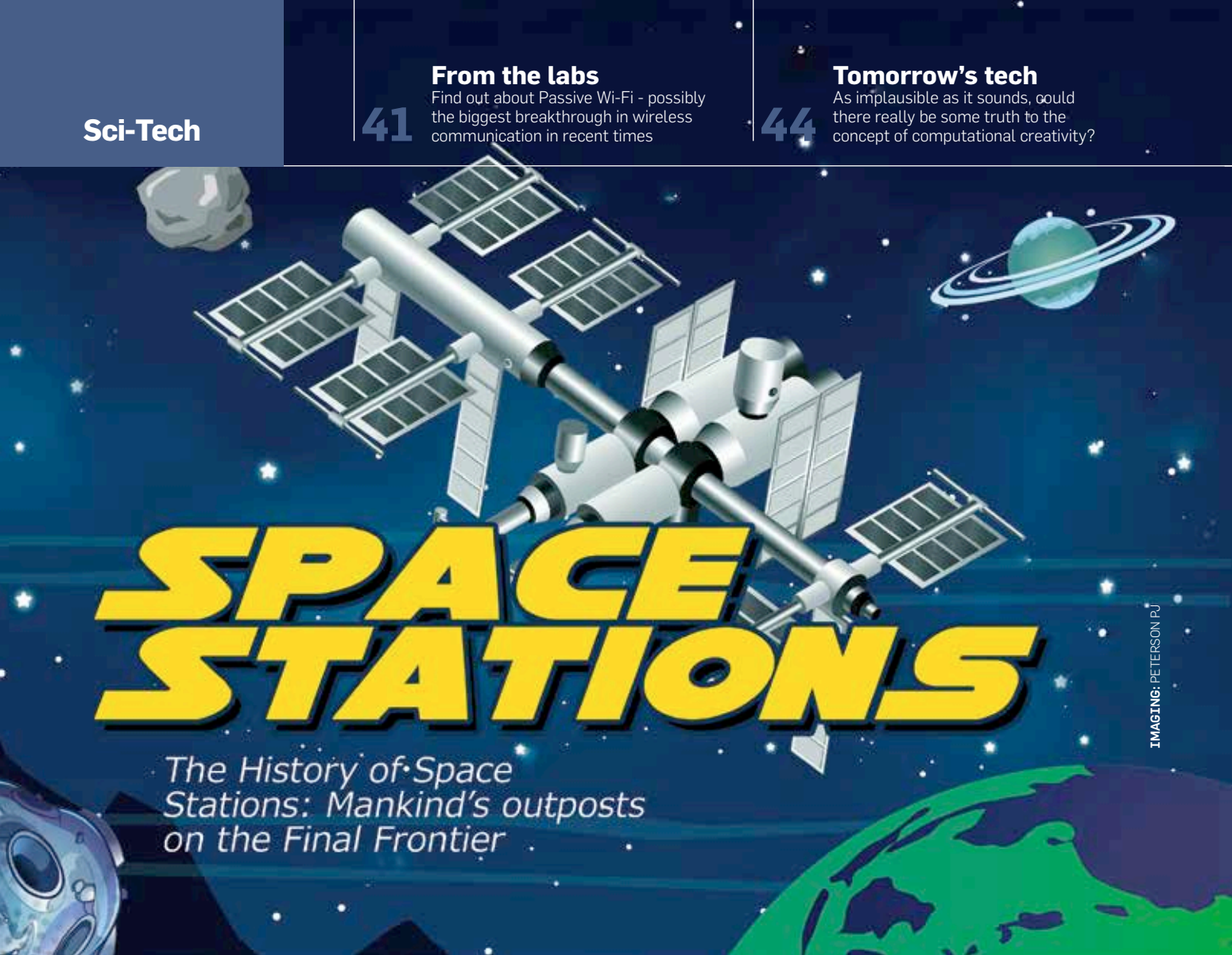


From the labs

Find out about Passive Wi-Fi - possibly the biggest breakthrough in wireless communication in recent times

Tomorrow's tech

As implausible as it sounds, could there really be some truth to the concept of computational creativity?



SPACE STATIONS

The History of Space Stations: Mankind's outposts on the Final Frontier

IMAGING: PETERSON PJ

Elroy Desmond
feedback@digit.in

This year, 20th February marked the 30th anniversary of the launch of the historic Mir Space Station. Owned and operated by Russia (then known as the Soviet Union), this low earth orbit space station orbited the earth for about 15 years until it was finally deorbited on 23rd March, 2001. For almost the entirety of its existence, Mir was the largest artificial satellite in orbit until it was succeeded by the International Space Station (ISS), which also broke Mir's record longest continuous human presence in space at 3,644 days. Mir still holds the record for the longest single human spaceflight, an honor that goes to Valeri Polyakov who spent 437 days and 18 hours on the station between 1994 and 1995.

The success of space stations like Mir marked a pivotal point in the mainstream success of space exploration programs.

Suddenly, every superpower wanted to put an astronaut on their own extraterrestrial home to help forward the progress of science and space exploration. In fact, the concept of the orbiting space station was first conceived as early as the 60's by scientists on both sides of the Cold War, who were fighting their own battles to decide who would be the first to rule the frontiers of outer space.

The Race to the first space station

Devised in part as a response to the the United States' successful Apollo program, a program that saw them place the first man on the moon, the Russians launched the Salyut space program, which aimed to launch and sustain the first manned orbiting space station. Under this program, they constructed and successfully launched the Salyut 1, on April 19, 1971. The Salyut 1 would go on to be the first of many stations under this pioneering program.

Featuring a monolithic design, Salyut 1 had several compartments that included three pressurized cabins, two of which were usable by the crew. Like all space stations that would come after it, accessing the Salyut 1 required the use of a docking spacecraft. For this the Russians used their Soyuz 7K-OKS spacecraft, which had already seen numerous successful launches. The launch of the Soyuz 10 on April 23rd, 1971, marked the first ever manned expedition to the Salyut 1, and the spacecraft comprised a crew of three Soviet cosmonauts (a Russian-coined term for astronaut). Unfortunately, their attempts to hard-dock the spacecraft were unsuccessful, and they were forced to abort the mission.

June 6th of the same year saw the Soyuz 11 perform the first successful attempt to dock a spacecraft on the Salyut 1. The Soyuz 11 flew 362 orbits in a duration of 23 days, but owing to technical problems on board the Salyut, the crew